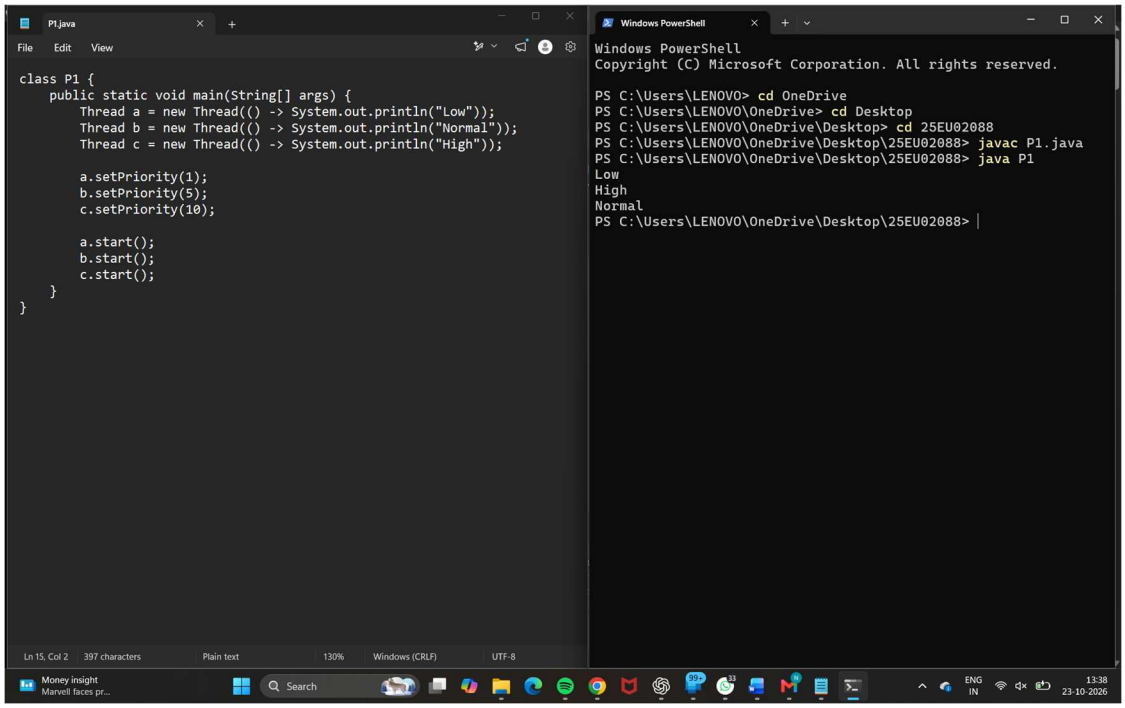
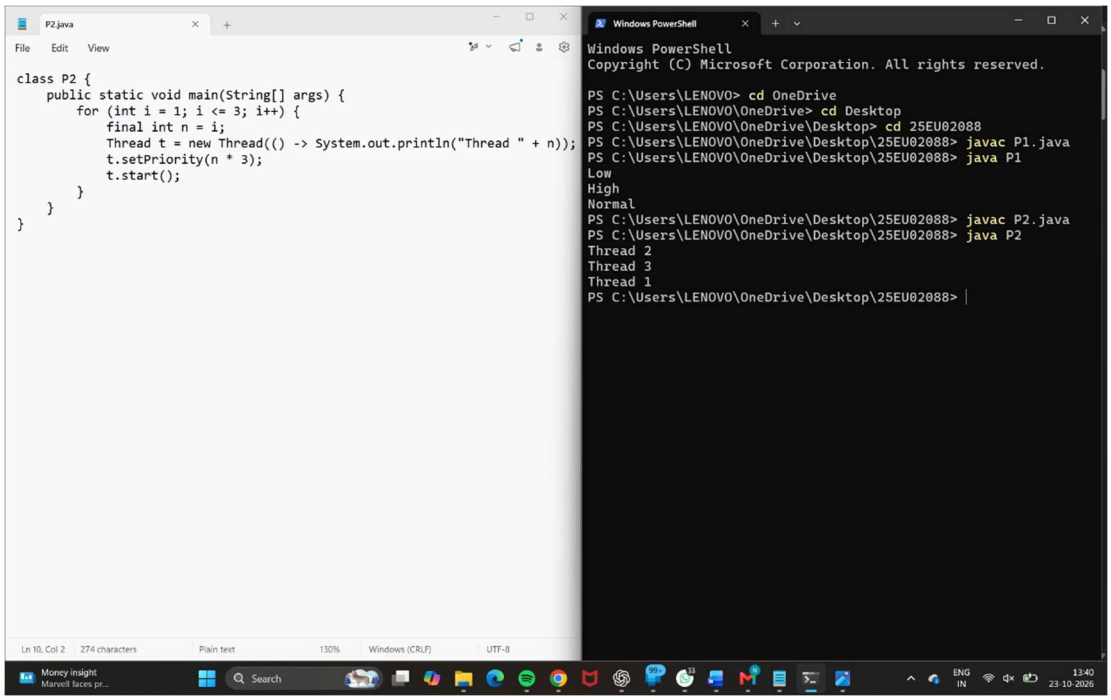
Multi threaded programming and Lambda expressions

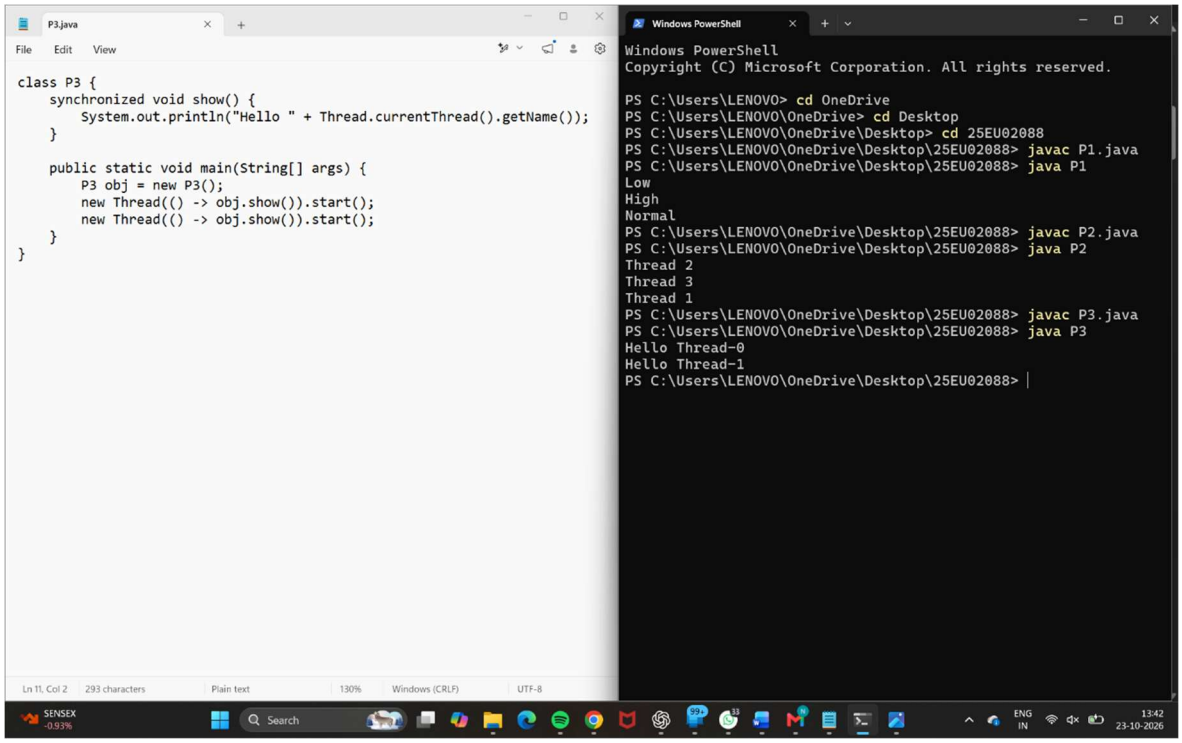
1.Write a Java program to create multiple threads and demonstrate different thread priorities using `MIN_PRIORITY`, `NORM_PRIORITY`, and `MAX_PRIORITY`.



2.Write a Java program to create three threads with different priorities and observe their execution order.



3. Write a Java program to demonstrate synchronization using a synchronized method for accessing a shared counter.



```
P3.java
File  Edit  View

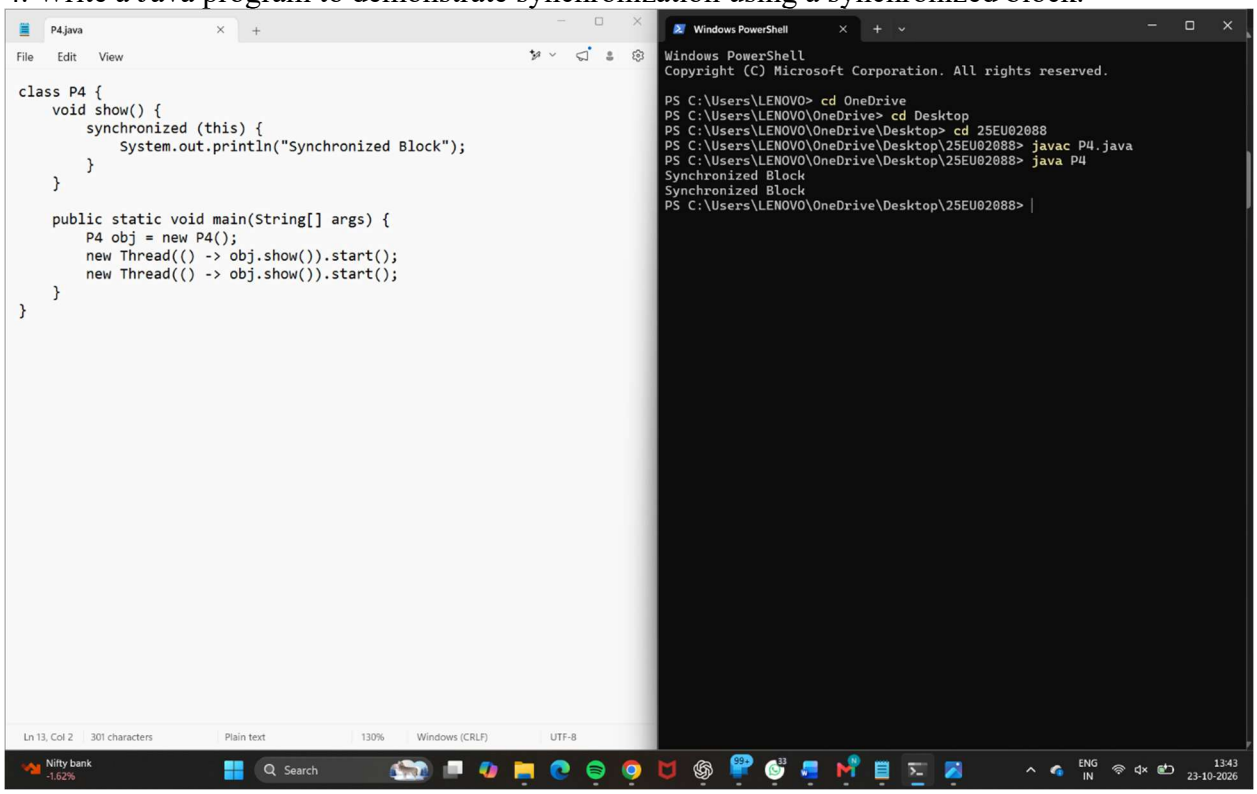
class P3 {
    synchronized void show() {
        System.out.println("Hello " + Thread.currentThread().getName());
    }

    public static void main(String[] args) {
        P3 obj = new P3();
        new Thread(() -> obj.show()).start();
        new Thread(() -> obj.show()).start();
    }
}
```

```
Windows PowerShell
Windows PowerShell
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PS C:\Users\LENOVO> cd OneDrive
PS C:\Users\LENOVO\OneDrive> cd Desktop
PS C:\Users\LENOVO\OneDrive\Desktop> cd 25EU02088
PS C:\Users\LENOVO\OneDrive\Desktop\25EU02088> javac P1.java
PS C:\Users\LENOVO\OneDrive\Desktop\25EU02088> java P1
Low
High
Normal
PS C:\Users\LENOVO\OneDrive\Desktop\25EU02088> javac P2.java
PS C:\Users\LENOVO\OneDrive\Desktop\25EU02088> java P2
Thread 2
Thread 3
Thread 1
PS C:\Users\LENOVO\OneDrive\Desktop\25EU02088> javac P3.java
PS C:\Users\LENOVO\OneDrive\Desktop\25EU02088> java P3
Hello Thread-0
Hello Thread-1
PS C:\Users\LENOVO\OneDrive\Desktop\25EU02088>
```

4. Write a Java program to demonstrate synchronization using a synchronized block.



```
P4.java
File  Edit  View

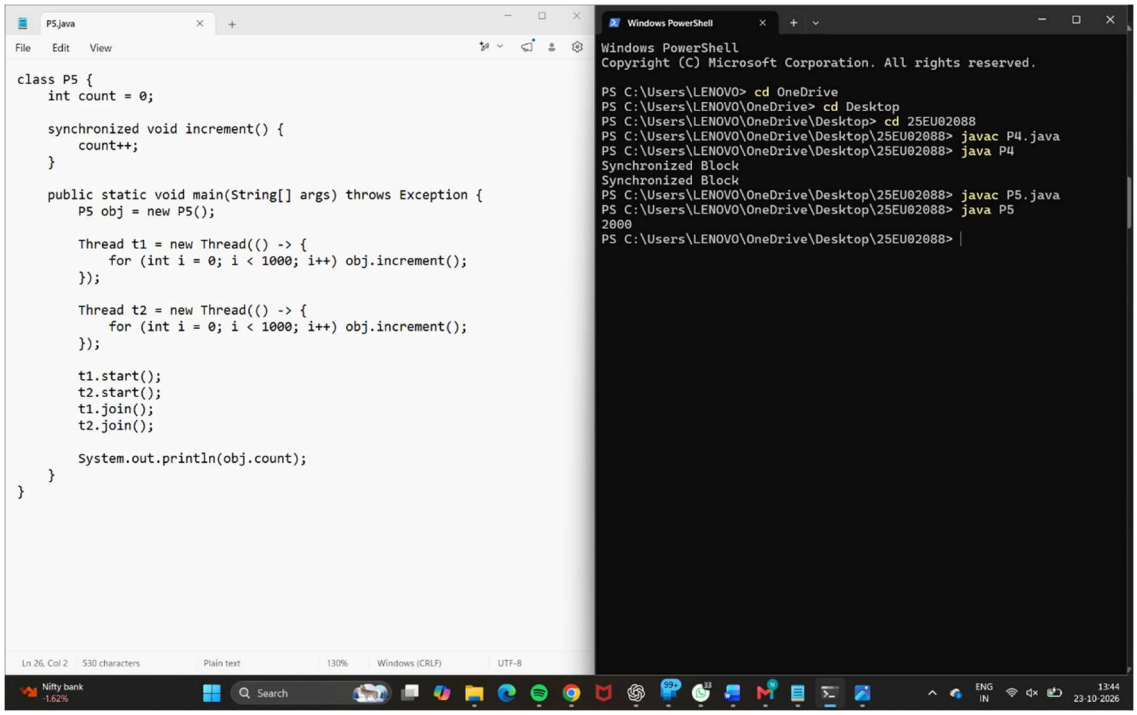
class P4 {
    void show() {
        synchronized (this) {
            System.out.println("Synchronized Block");
        }
    }

    public static void main(String[] args) {
        P4 obj = new P4();
        new Thread(() -> obj.show()).start();
        new Thread(() -> obj.show()).start();
    }
}
```

```
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PS C:\Users\LENOVO\OneDrive> cd Desktop
PS C:\Users\LENOVO\OneDrive\Desktop> cd 25EU02088
PS C:\Users\LENOVO\OneDrive\Desktop\25EU02088> javac P4.java
PS C:\Users\LENOVO\OneDrive\Desktop\25EU02088> java P4
Synchronized Block
Synchronized Block
PS C:\Users\LENOVO\OneDrive\Desktop\25EU02088>
```

5. Write a Java program to demonstrate a race condition and eliminate it using synchronization.



```
P5.java
File  Edit  View

class P5 {
    int count = 0;

    synchronized void increment() {
        count++;
    }

    public static void main(String[] args) throws Exception {
        P5 obj = new P5();

        Thread t1 = new Thread(() -> {
            for (int i = 0; i < 1000; i++) obj.increment();
        });

        Thread t2 = new Thread(() -> {
            for (int i = 0; i < 1000; i++) obj.increment();
        });

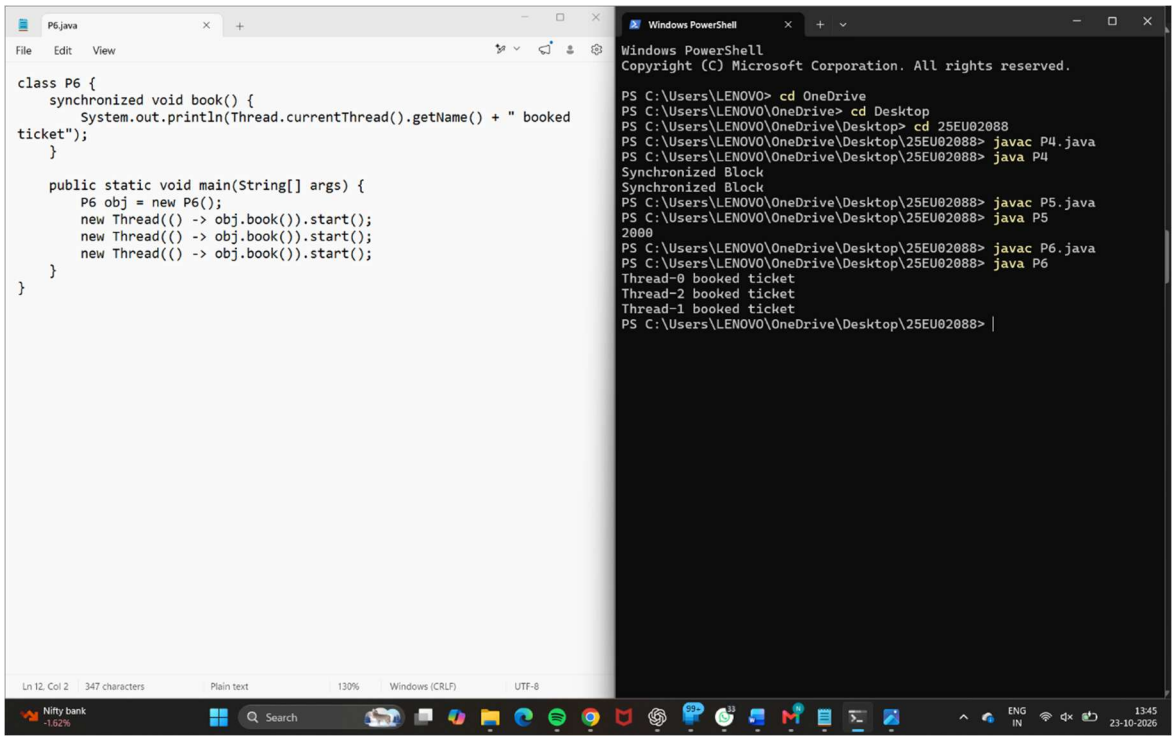
        t1.start();
        t2.start();
        t1.join();
        t2.join();

        System.out.println(obj.count);
    }
}
```

```
Windows PowerShell
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PS C:\Users\LENOVO> cd OneDrive
PS C:\Users\LENOVO\OneDrive> cd Desktop
PS C:\Users\LENOVO\OneDrive\Desktop> cd 25EU02088
PS C:\Users\LENOVO\OneDrive\Desktop\25EU02088> javac P4.java
PS C:\Users\LENOVO\OneDrive\Desktop\25EU02088> java P4
Synchronized Block
Synchronized Block
PS C:\Users\LENOVO\OneDrive\Desktop\25EU02088> javac P5.java
PS C:\Users\LENOVO\OneDrive\Desktop\25EU02088> java P5
2000
PS C:\Users\LENOVO\OneDrive\Desktop\25EU02088>
```

6. Write a Java program to implement a synchronized ticket-booking system where multiple threads attempt to book tickets simultaneously.



```
P6.java
File  Edit  View

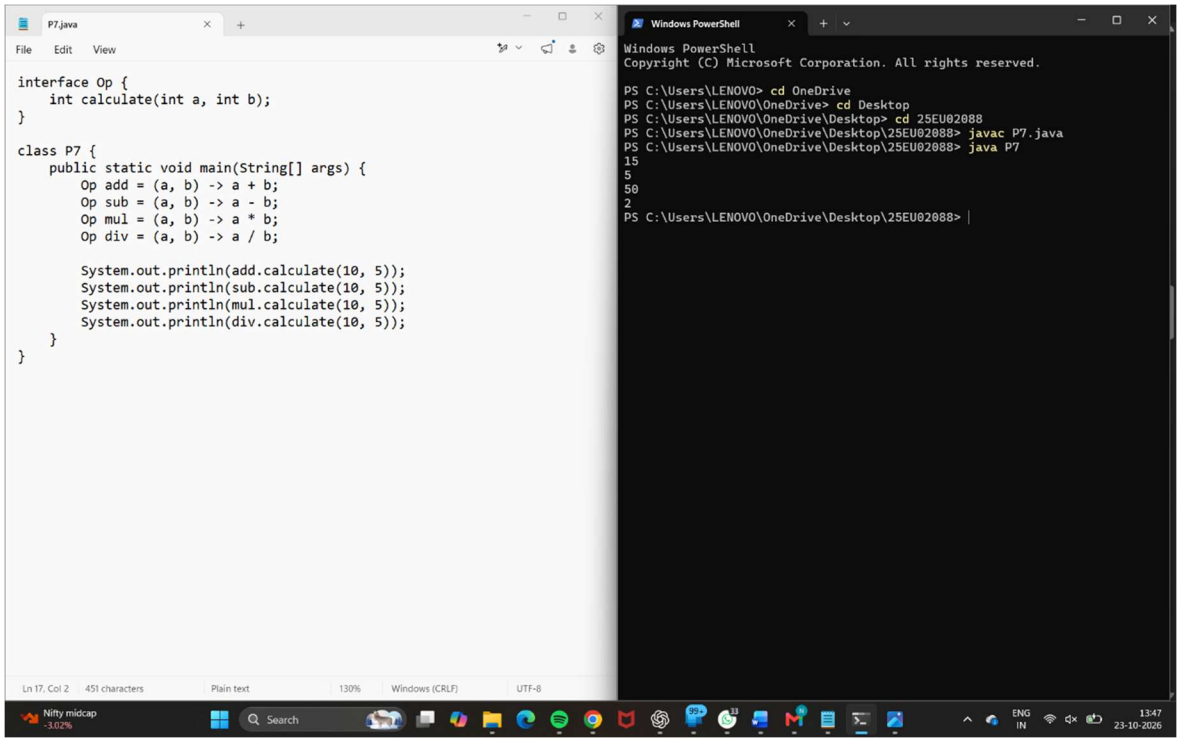
class P6 {
    synchronized void book() {
        System.out.println(Thread.currentThread().getName() + " booked
ticket");
    }

    public static void main(String[] args) {
        P6 obj = new P6();
        new Thread(() -> obj.book()).start();
        new Thread(() -> obj.book()).start();
        new Thread(() -> obj.book()).start();
    }
}
```

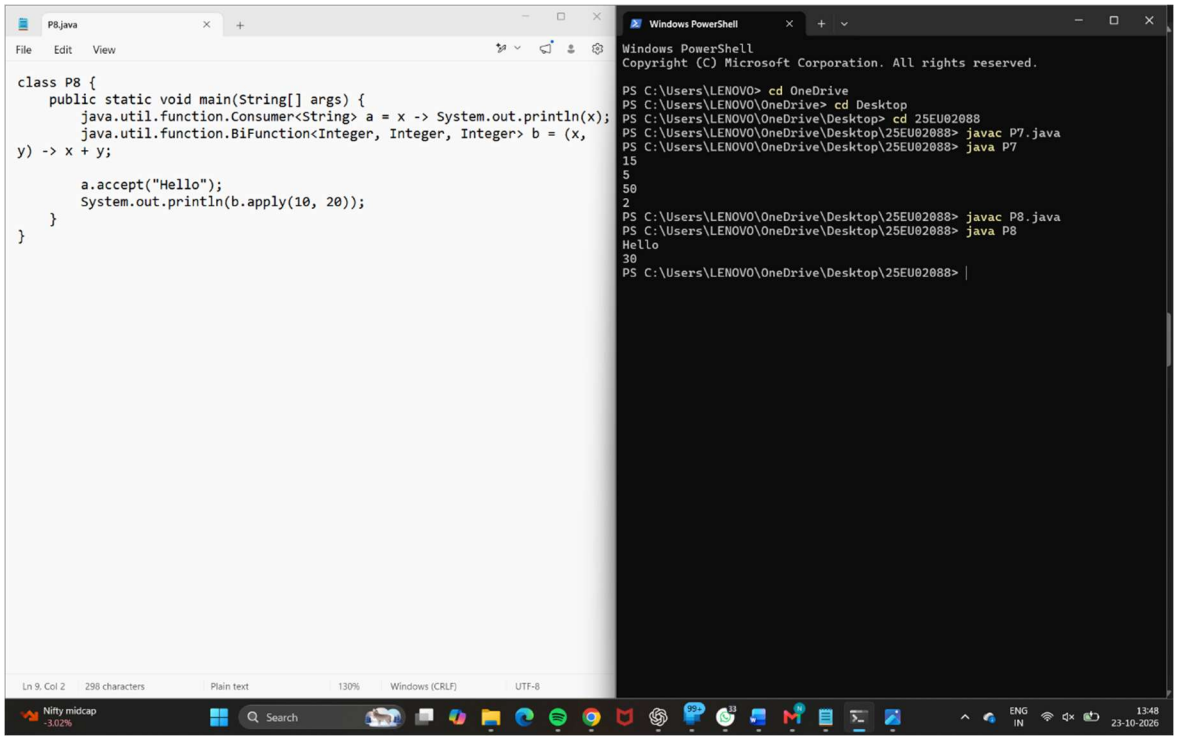
```
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PS C:\Users\LENOVO> cd OneDrive
PS C:\Users\LENOVO\OneDrive> cd Desktop
PS C:\Users\LENOVO\OneDrive\Desktop> cd 25EU02088
PS C:\Users\LENOVO\OneDrive\Desktop\25EU02088> javac P4.java
PS C:\Users\LENOVO\OneDrive\Desktop\25EU02088> java P4
Synchronized Block
Synchronized Block
PS C:\Users\LENOVO\OneDrive\Desktop\25EU02088> javac P5.java
PS C:\Users\LENOVO\OneDrive\Desktop\25EU02088> java P5
2000
PS C:\Users\LENOVO\OneDrive\Desktop\25EU02088> javac P6.java
PS C:\Users\LENOVO\OneDrive\Desktop\25EU02088> java P6
Thread-0 booked ticket
Thread-2 booked ticket
Thread-1 booked ticket
PS C:\Users\LENOVO\OneDrive\Desktop\25EU02088>
```

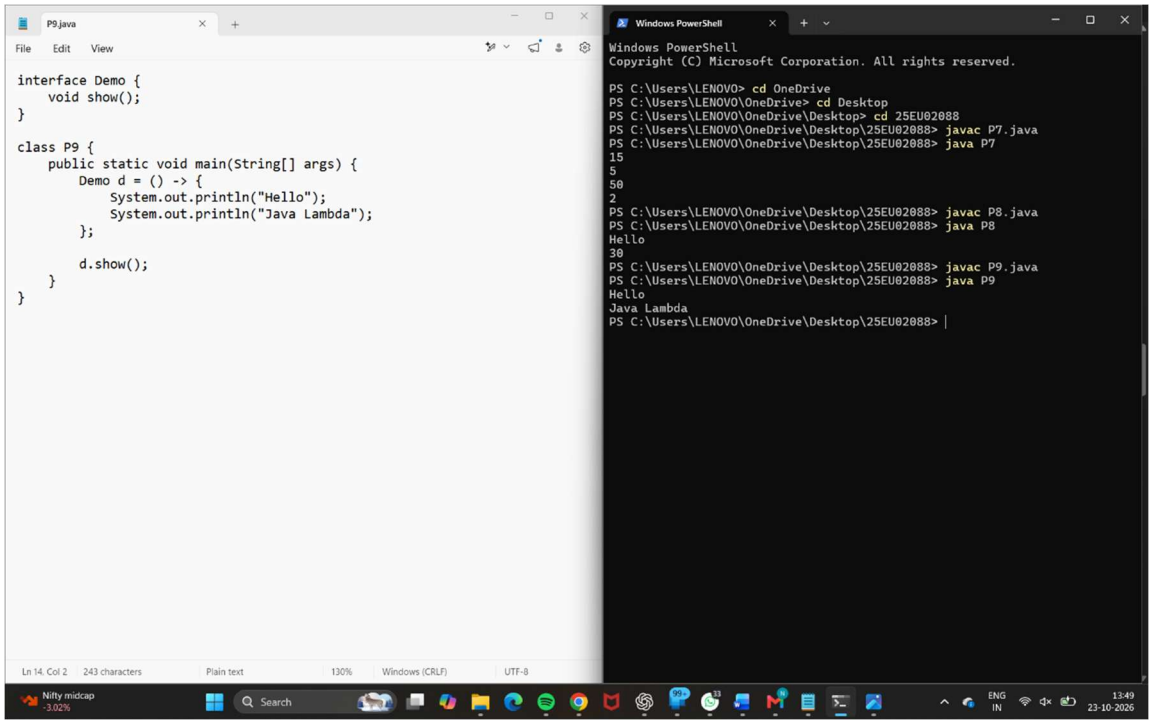
7. Write a Java program to introduce lambda expressions by implementing a functional interface for addition, subtraction, multiplication, and division.



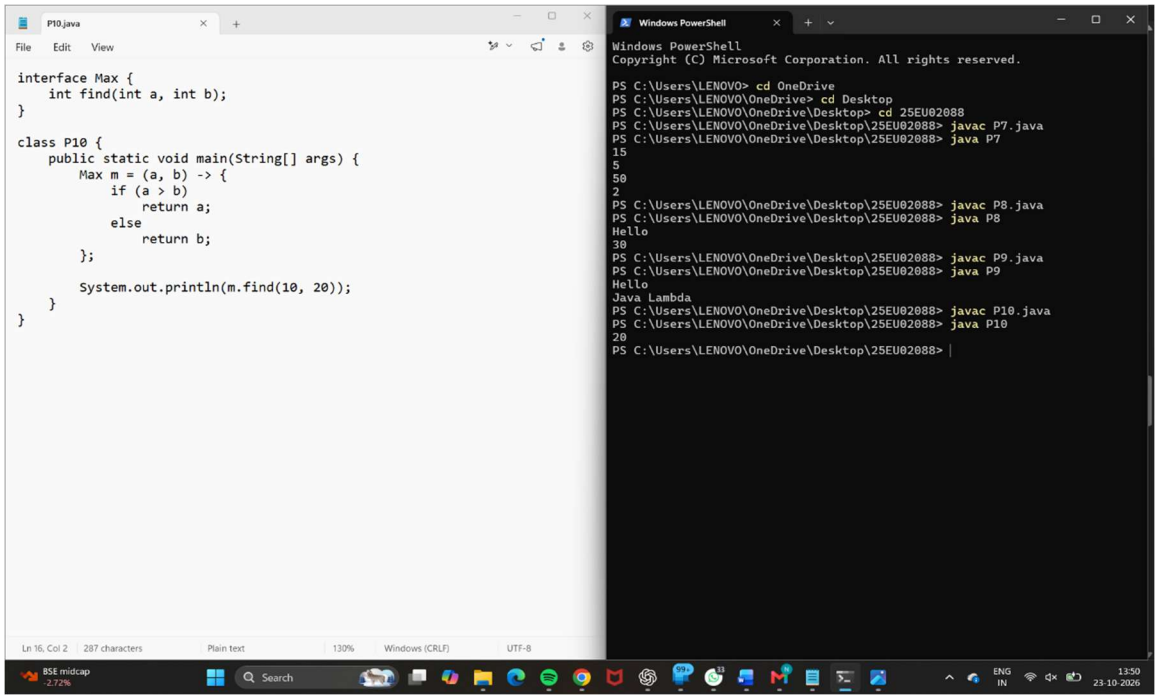
8. Write a Java program to demonstrate a lambda expression with a single parameter and multiple parameters.



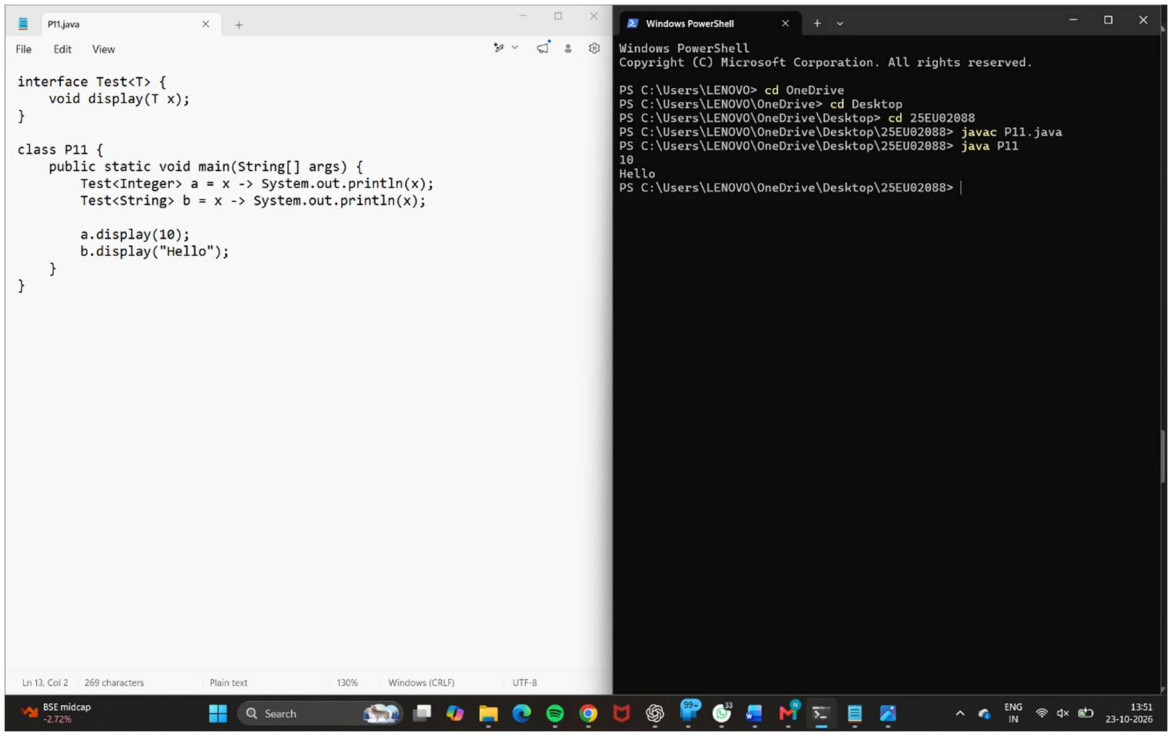
9. Write a Java program to demonstrate a block lambda expression containing multiple statements.



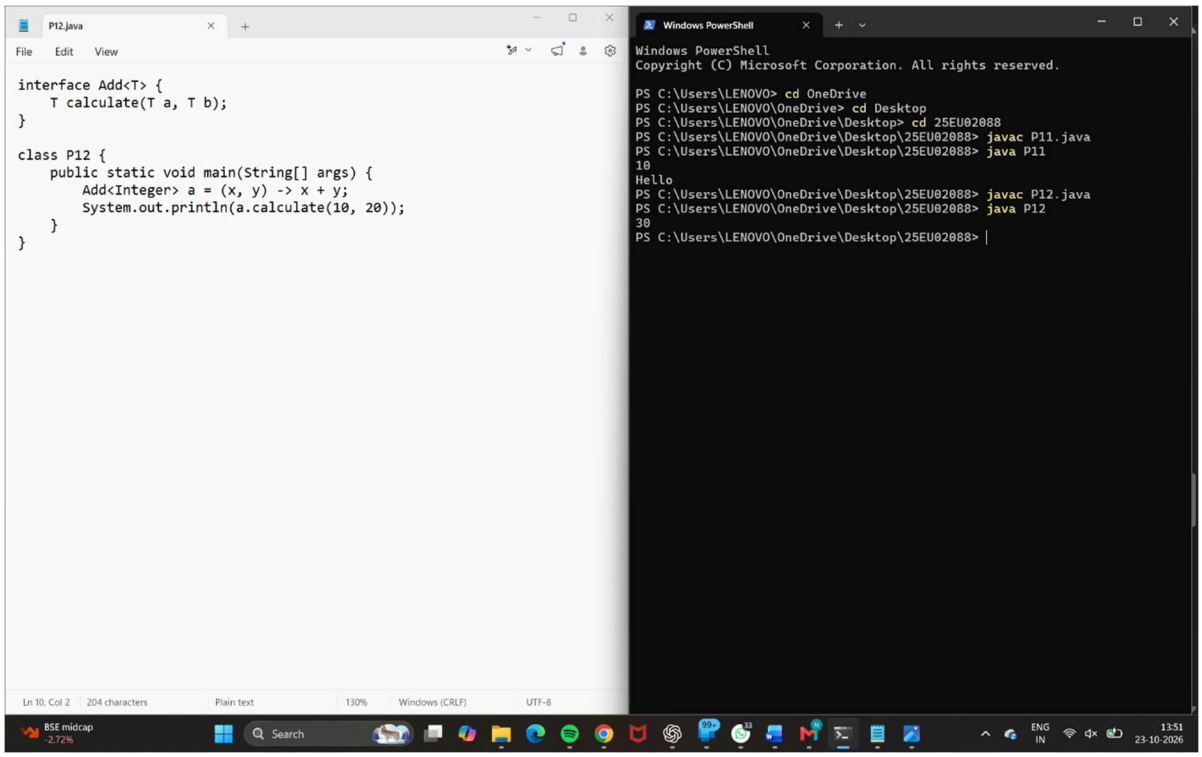
10. Write a Java program using a block lambda expression to find the largest of two numbers and return the result.



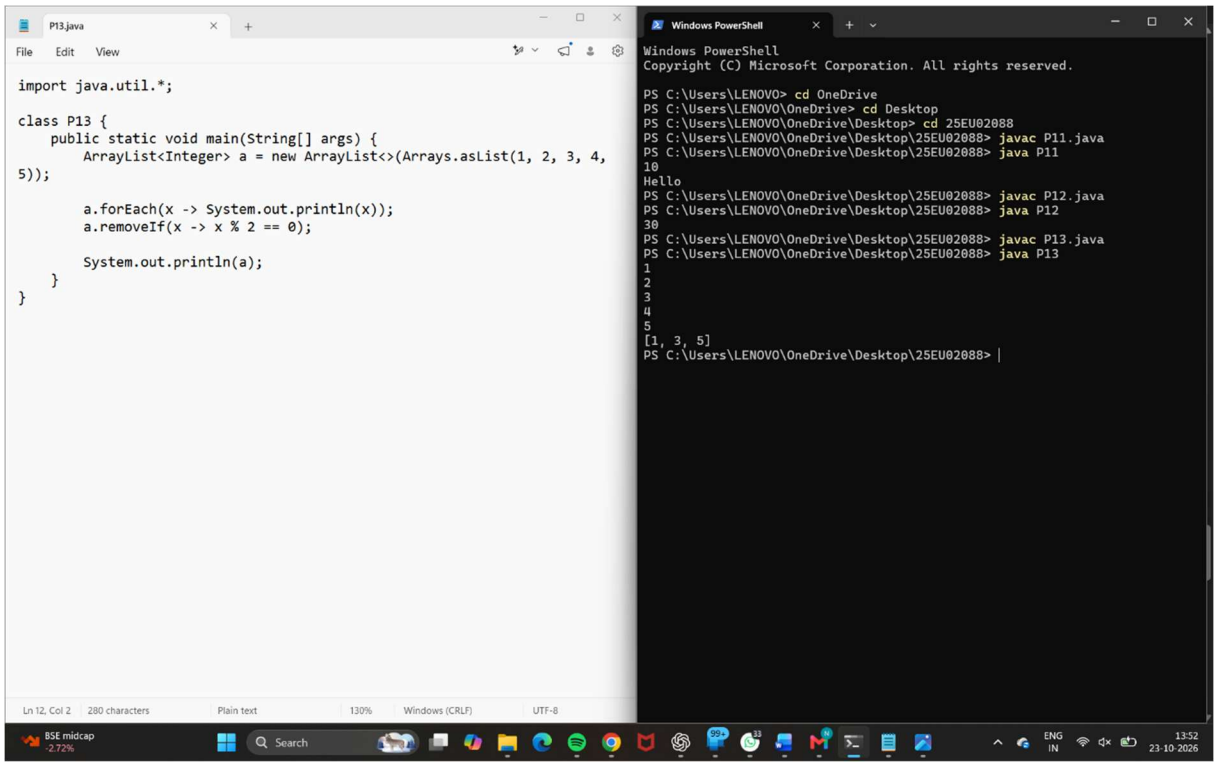
11. Write a Java program to create a generic functional interface and implement it using lambda expressions for different data types.



12. Write a Java program to create a generic functional interface that accepts two values and performs an operation using a lambda expression.



13. Write a Java program to use lambda expressions to display and filter elements of an `ArrayList`.



```
P13.java
File  Edit  View

import java.util.*;

class P13 {
    public static void main(String[] args) {
        ArrayList<Integer> a = new ArrayList<>(Arrays.asList(1, 2, 3, 4,
5));

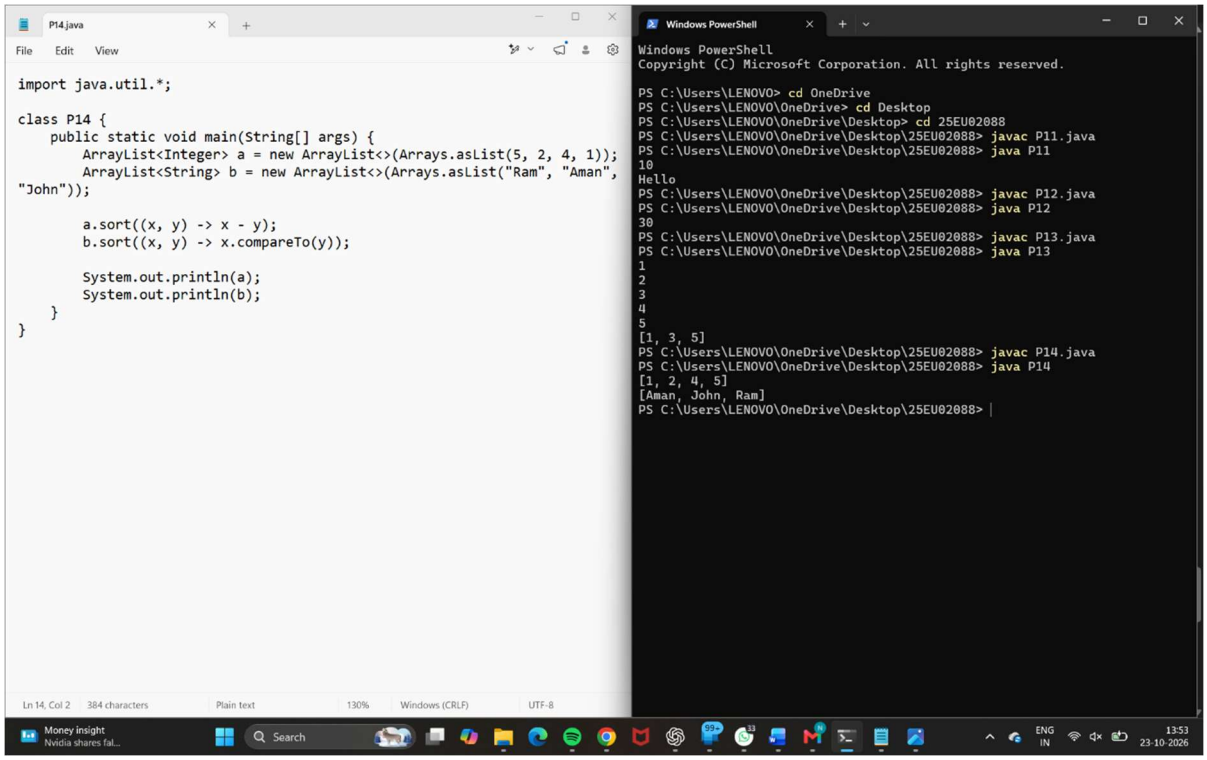
        a.forEach(x -> System.out.println(x));
        a.removeIf(x -> x % 2 == 0);

        System.out.println(a);
    }
}
```

```
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PS C:\Users\LENOVO> cd OneDrive
PS C:\Users\LENOVO\OneDrive> cd Desktop
PS C:\Users\LENOVO\OneDrive\Desktop> cd 25EU02088
PS C:\Users\LENOVO\OneDrive\Desktop\25EU02088> javac P11.java
PS C:\Users\LENOVO\OneDrive\Desktop\25EU02088> java P11
10
Hello
PS C:\Users\LENOVO\OneDrive\Desktop\25EU02088> javac P12.java
PS C:\Users\LENOVO\OneDrive\Desktop\25EU02088> java P12
30
PS C:\Users\LENOVO\OneDrive\Desktop\25EU02088> javac P13.java
PS C:\Users\LENOVO\OneDrive\Desktop\25EU02088> java P13
1
2
3
4
5
[1, 3, 5]
PS C:\Users\LENOVO\OneDrive\Desktop\25EU02088>
```

14. Write a Java program to sort a list of integers and strings using lambda expressions.



```
P14.java
File  Edit  View

import java.util.*;

class P14 {
    public static void main(String[] args) {
        ArrayList<Integer> a = new ArrayList<>(Arrays.asList(5, 2, 4, 1));
        ArrayList<String> b = new ArrayList<>(Arrays.asList("Ram", "Aman",
"John"));

        a.sort((x, y) -> x - y);
        b.sort((x, y) -> x.compareTo(y));

        System.out.println(a);
        System.out.println(b);
    }
}
```

```
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PS C:\Users\LENOVO> cd OneDrive
PS C:\Users\LENOVO\OneDrive> cd Desktop
PS C:\Users\LENOVO\OneDrive\Desktop> cd 25EU02088
PS C:\Users\LENOVO\OneDrive\Desktop\25EU02088> javac P11.java
PS C:\Users\LENOVO\OneDrive\Desktop\25EU02088> java P11
10
Hello
PS C:\Users\LENOVO\OneDrive\Desktop\25EU02088> javac P12.java
PS C:\Users\LENOVO\OneDrive\Desktop\25EU02088> java P12
30
PS C:\Users\LENOVO\OneDrive\Desktop\25EU02088> javac P13.java
PS C:\Users\LENOVO\OneDrive\Desktop\25EU02088> java P13
1
2
3
4
5
[1, 3, 5]
PS C:\Users\LENOVO\OneDrive\Desktop\25EU02088> javac P14.java
PS C:\Users\LENOVO\OneDrive\Desktop\25EU02088> java P14
[1, 2, 4, 5]
[Aman, John, Ram]
PS C:\Users\LENOVO\OneDrive\Desktop\25EU02088>
```

15. Write a Java program to create and execute a thread using a lambda expression.

